2023/24

Solution Mid Term Exam - Applied Maths
Part 2

Exercise 1

$$x^4 - 6x^2 + \varepsilon x + 8 = 0 \qquad (1)$$

$$\varepsilon << 1$$

$$x \sim x_0 + \varepsilon x_1 + O(\varepsilon^2)$$

$$\Rightarrow x^4 \sim x_0^4 + 4x_0^3 \varepsilon x_1 + O(\varepsilon^2)$$

$$x^2 \sim x_0^2 + 2x_0 x_1 \varepsilon + O(\varepsilon^2)$$

$$(1) \Rightarrow x_0^4 + 4x_0^3 x_1 \varepsilon - 6x_0^2 - 12 x_0 x_1 \varepsilon + \varepsilon x_0 + 8 + O(\varepsilon^2) = 0$$

$$\Rightarrow \begin{cases} x_0^4 - 6x_0^2 + 8 = 0 & (i) \\ 4x_0^3 x_1 - 12 x_0 x_1 + x_0 = 0 & (ii) \end{cases}$$

(i) We set: $X_0 = x_0^2$ then $X_0^2 - 6X_0 + 8 = 0$

$$\Delta = 36 - 32 = 4$$

Then: $X_0 = \dfrac{6 \pm 2}{2} = \begin{cases} 4 \\ \text{or} \\ 2 \end{cases}$

Then: $x_0 = \pm\sqrt{X_0} = \pm 2$ ou $\pm\sqrt{2}$

(ii) gives: $x_1 = \dfrac{1}{12 - 4x_0^2}$

So if $x_0 = \pm 2$, $x_1 = \dfrac{1}{12 - 4\times 4} = -\dfrac{1}{4}$

if $x_0 = \pm\sqrt{2}$, $x_1 = \dfrac{1}{12 - 4\times 2} = \dfrac{1}{4}$

Thus the 4 roots at order 1 in ε are:

$$\begin{cases} x \underset{\varepsilon \to 0}{\sim} 2 - \dfrac{\varepsilon}{4} \\ x \underset{\varepsilon \to 0}{\sim} -2 - \dfrac{\varepsilon}{4} \\ x \underset{\varepsilon \to 0}{\sim} \sqrt{2} + \dfrac{\varepsilon}{4} \\ x \underset{\varepsilon \to 0}{\sim} -\sqrt{2} + \dfrac{\varepsilon}{4} \end{cases}$$

Exercise 2

$$\varepsilon x^3 + x^2 + x + 2\varepsilon = 0 \qquad (2)$$

1) This equation is singular because the highest degree term is multiplied by ε. The there is a singularity when $\varepsilon \to 0$: if $\varepsilon = 0$ the equation is of degree 2 giving 2 regular roots whereas if $\varepsilon \ll 1$, $\varepsilon \neq 0$ it has 3 roots. The third root is thus a singular root.

this singular root must be very large when ε is very small such that the first term stays finite as $\varepsilon \to 0$.

2) if $x \sim x_0 + x_1\varepsilon + x_2\varepsilon^2 + O(\varepsilon^3)$

then $x^3 \sim x_0^3 + 3x_0^2x_1\varepsilon + (3x_0^2x_2 + 3x_0x_1^2)\varepsilon^2 + O(\varepsilon^3)$

and $x^2 \sim x_0^2 + 2x_0x_1\varepsilon + (x_1^2 + 2x_0x_2)\varepsilon^2 + O(\varepsilon^3)$

$$(2) \Rightarrow \varepsilon x_0^3 + 3x_0^2x_1\varepsilon^2 + x_0^2 + 2x_0x_1\varepsilon + (x_1^2 + 2x_0x_2)\varepsilon^2 + x_0 + x_1\varepsilon + x_2\varepsilon^2 + 2\varepsilon = 0$$

$$\Rightarrow \begin{cases} \varepsilon^0 : & x_0^2 + x_0 = 0 \\ \varepsilon^1 : & x_0^3 + 2x_0x_1 + x_1 + 2 = 0 \\ \varepsilon^2 : & 3x_0^2x_1 + x_1^2 + 2x_0x_2 + x_2 = 0 \end{cases}$$

Thus $x_0 = 0$ or $x_0 = -1$

with $x_1 = \dfrac{-2 - x_0^3}{2x_0 + 1}$

and $x_2 = \dfrac{-x_1^2 - 3x_0^2 x_1}{2x_0 + 1}$

We obtain $x_0 = 0$, $x_1 = -2$, $x_2 = -4$

or

$x_0 = -1$, $x_1 = 1$, $x_2 = 4$

And the 2 regular roots of order 2 in ε are

$$\begin{cases} x \sim -2\varepsilon - 4\varepsilon^2 \\ x \sim -1 + \varepsilon + 4\varepsilon^2 \end{cases}$$

3) Let $x = \dfrac{X}{\delta}$ to obtain the third root

(if is expected to tend to α as $\varepsilon \to 0$)

Then

$$(2) \Rightarrow \varepsilon \frac{X^3}{\delta^3} + \frac{X^2}{\delta^2} + \frac{X}{\delta} + 2\varepsilon = 0$$

$$\Rightarrow \frac{\varepsilon}{\delta^3} X^3 + \frac{1}{\delta^2} X^2 + \frac{1}{\delta} X + 2\varepsilon = 0$$

① ② ③ ④

- ① ~ ② $\Rightarrow \frac{\varepsilon}{\delta^3} \sim \frac{1}{\delta^2} \Rightarrow \varepsilon \sim \delta$

  then ① ~ ② ~ $\frac{1}{\varepsilon^2} \gg$ ③ ~ $\frac{1}{\varepsilon}$ and ④ ~ ε

  Ok Dominant balance

- ① ~ ③ $\Rightarrow \frac{\varepsilon}{\delta^3} \sim \frac{1}{\delta} \Rightarrow \varepsilon \sim \delta^2$

  then ① ~ ③ ~ $\frac{1}{\varepsilon^2}$ but ② ~ $\frac{1}{\delta^2} \sim \frac{1}{\varepsilon^2}$ same order,

  thus No dominant balance

- ① ~ ④ $\Rightarrow \delta \sim 1$ not ok (same as regular)
- ② ~ ③ $\Rightarrow \delta \sim 1$ same
- ② ~ ④ $\Rightarrow \frac{1}{\delta^2} \sim \varepsilon$ not possible with $\delta << 1$
- ③ ~ ④ $\Rightarrow \frac{1}{\delta} \sim \varepsilon$ same

Thus the only case to be considered here is $\delta \sim \varepsilon$ to obtain the singular root.

4) with $\varepsilon \sim \delta$

$$(2) \Rightarrow \frac{1}{\delta^2} x^3 + \frac{1}{\delta^2} x^2 + \frac{1}{\delta} x + 2\delta = 0$$

$$\Rightarrow x^3 + x^2 + \delta x + 2\delta^3 = 0$$

$$x = x_0 + x_1 \delta + x_2 \delta^2 + O(\delta^3)$$

$$\Rightarrow x^3 \sim x_0^3 + 3x_0^2 x_1 \delta + \left(3x_0^2 x_2 + 3x_0 x_1^2\right)\delta^2 + O(\delta^3)$$

$$x^2 \sim x_0^2 + 2x_0 x_1 \delta + \left(x_1^2 + 2x_0 x_2\right)\delta^2 + O(\delta^3)$$

thus

at δ^0: $x_0^3 + x_0^2 = 0 \Rightarrow x_0 = 0$ or $x_0 = -1$

We expect $x_0 \neq 0$ because if $x_0 = 0$ the root is regular.

thus $x_0 = -1$

at order δ^1: with $x_0 \neq 0$

$3x_0^2 x_1 + 2x_0 x_1 + x_0 = 0$ thus $x_1 = \dfrac{-x_0}{3x_0^2 + 2x_0}$

and at order δ^2 : with $X_0 \neq 0$

$$3X_0^2X_2 + 3X_0X_1^2 + X_1^2 + 2X_0X_2 + X_1 = 0$$

which gives :

$$X_2 = \frac{-X_1 - 3X_0X_1^2 - X_1^2}{3X_0^2 + 2X_0}$$

We then obtain :

$$\begin{cases} X_0 = -1 \\ X_1 = \dfrac{1}{3-2} = 1 \\ X_2 = \dfrac{-1 - 3(-1)1^2 - 1^2}{3(-1)^2 + 2(-1)} = 1 \end{cases}$$

Thus $x = -1 + \delta + \delta^2 + O(\delta^3)$

and with $x = \dfrac{X}{\delta}$ and $\delta \sim \varepsilon$

we obtain : $x \sim -\dfrac{1}{\varepsilon} + 1 + \varepsilon$

$\varepsilon \to 0$ ε

which is the singular root ($x \underset{\varepsilon \to 0}{\longrightarrow} -\infty$)